



MANIPAL

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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5153 Linux OS and Scripting – 2_Lab

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25/08/2026

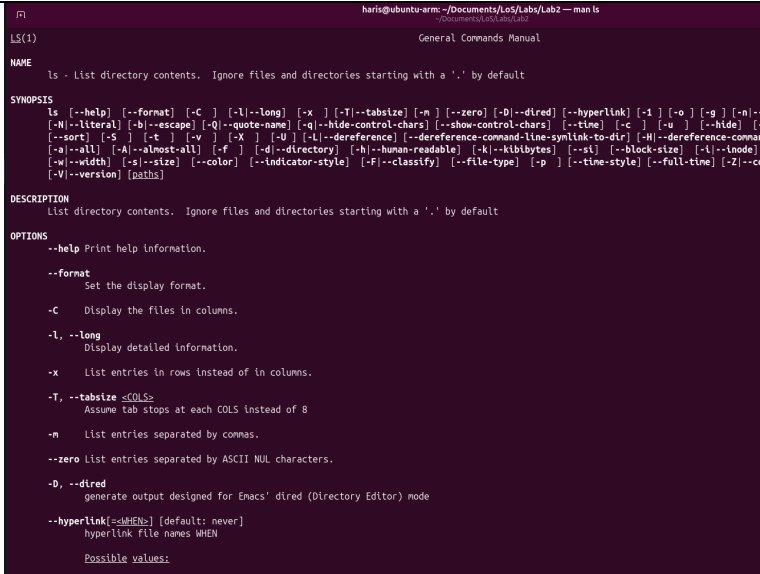
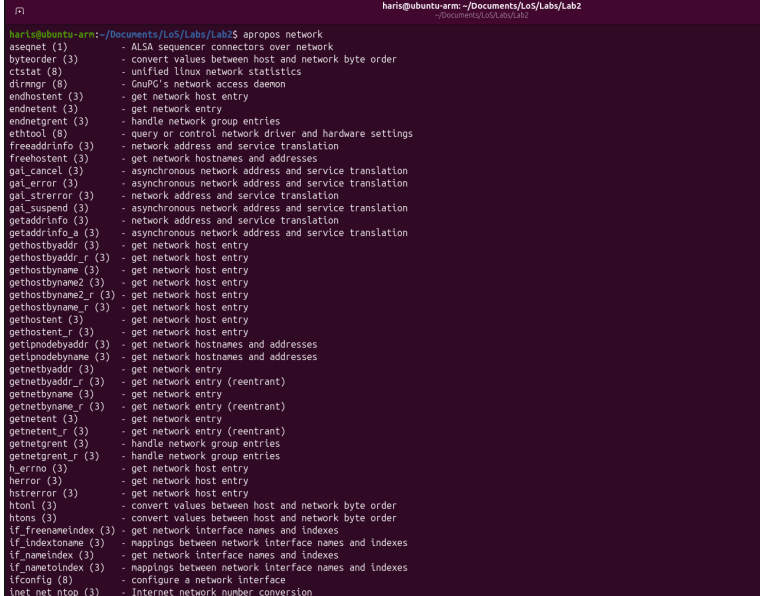
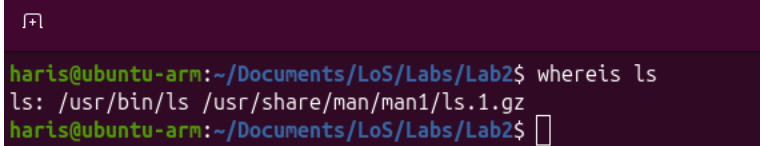


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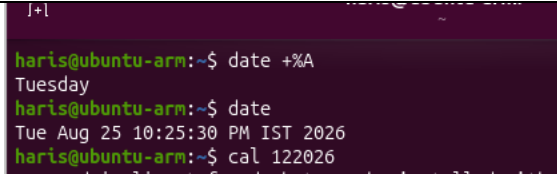
1. Getting Help

Task	Com mand Name	Synt ax	Exa mple	Screenshots
To get manual page for the known command	man	man [options] [section] command	man 1 ls	 <pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ man 1 ls LS(1) General Commands Manual NAME ls - List directory contents. Ignore files and directories starting with a '.' by default SYNOPSIS ls [-help] [--format] [-C] [-l --long] [-x] [-T --tabsize] [-n] [--zero] [-D --dired] [--hyperlink] [-1] [-o] [-g] [-n] [-N --literal] [-b --escape] [-Q --quote-name] [-q --hide-control-chars] [--show-control-chars] [--time] [-c] [-u] [--hide] [- [--sort] [-S] [-t] [-v] [-X] [-d] [-l --dereference] [--dereference-command-line-synlink-to-dir] [-H --dereference-comma -a --all] [-A --almost-all] [-f] [-d --directory] [-b --human-readable] [-k --kibibytes] [--si] [--block-size] [-l --inode] [-w --width] [-s --size] [--color] [--indicator-style] [-F --classify] [--file-type] [-p] [--time-style] [--full-time] [-Z --co [-V --version] [paths] DESCRIPTION List directory contents. Ignore files and directories starting with a '.' by default OPTIONS --help Print help information. --format Set the display format. -C Display the files in columns. -l, --long Display detailed information. -x List entries in rows instead of in columns. -T, --tabsize <COLS> Assume tab stops at each COLS instead of 8 -m List entries separated by commas. --zero List entries separated by ASCII NUL characters. -D, --dired generate output designed for Emacs' dired (Directory Editor) mode --hyperlink[=style] [default: never] hyperlink file names WHEN Possible values: </pre>
To get manual page for the unknown command	apropos	apropos keyword	apropos network	 <pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ apropos network aseqnet (1) - ALSA sequencer connectors over network byteorder (3) - convert values between host and network byte order ctstat (8) - unified Linux network statistics dirmngr (8) - GnuPG's network access daemon endhostent (3) - get network host entry endnetent (3) - get network entry endnetgrent (3) - handle network group entries ethtool (8) - query or control network driver and hardware settings freeddrinfo (3) - network address and service translation freehostent (3) - get network hostnames and addresses gai_cancel (3) - asynchronous network address and service translation gai_error (3) - asynchronous network address and service translation gai_strerror (3) - network address and service translation gai_suspend (3) - asynchronous network address and service translation getaddrinfo (3) - network address and service translation getaddrinfo_a (3) - asynchronous network address and service translation gethostbyname (3) - get network host entry gethostbyname_r (3) - get network host entry gethostbyname2 (3) - get network host entry gethostbyname2_r (3) - get network host entry gethostent (3) - get network host entry gethostent_r (3) - get network host entry getipnodebyaddr (3) - get network hostnames and addresses getipnodebyname (3) - get network hostnames and addresses getnetbyaddr (3) - get network entry getnetbyaddr_r (3) - get network entry (reentrant) getnetbyname (3) - get network entry getnetbyname_r (3) - get network entry (reentrant) getnetent (3) - get network entry getnetent_r (3) - get network entry (reentrant) getnetgrent (3) - handle network group entries getnetgrent_r (3) - handle network group entries h_errno (3) - get network host entry herror (3) - get network host entry hstrerror (3) - get network host entry htonl (3) - convert values between host and network byte order htons (3) - convert values between host and network byte order if_freenameindex (3) - get network interface names and indexes if_indexname (3) - mappings between network interface names and indexes if_nameindex (3) - get network interface names and indexes if_nameindex (3) - mappings between network interface names and indexes ifconfig (8) - configure a network interface inet_net_ntop (3) - Internet network number conversion </pre>
To know the source file binary	whereis	whereis [options] command	whereis ls	 <pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ whereis ls ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ </pre>

To know the path of the command	which	which command	which ls	<pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ which ls /usr/bin/ls haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ █ </pre>
To know the command and is external or internal	type	type command	type cd	<pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ type cd cd is a shell builtin haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ █ </pre>
To get help for the internal command	help	help command	help echo	<pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ help cd cd: cd [-L [-P [-e]]] [-@] [dir] Change the shell working directory. Change the current directory to DIR. The default DIR is the value of the HOME shell variable. If DIR is "-", it is converted to \$OLDPWD. The variable CDPATH defines the search path for the directory containing DIR. Alternative directory names in CDPATH are separated by a colon (:). A null directory name is the same as the current directory. If DIR begins with a slash (/), then CDPATH is not used. If the directory is not found, and the shell option 'cdable_vars' is set, the word is assumed to be a variable name. If that variable has a value, its value is used for DIR. Options: -L force symbolic links to be followed: resolve symbolic links in DIR after processing instances of '..' -P use the physical directory structure without following symbolic links: resolve symbolic links in DIR before processing instances of '..' -e if the -P option is supplied, and the current working directory cannot be determined successfully, exit with a non-zero status -@ on systems that support it, present a file with extended attributes as a directory containing the file attributes The default is to follow symbolic links, as if '-L' were specified. '..' is processed by removing the immediately previous pathname component back to a slash or the beginning of DIR. Exit Status: Returns 0 if the directory is changed, and if \$PWD is set successfully when -P is used; non-zero otherwise. haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ █ </pre>

To list out bash commands	comp gen	comp gen - flag	comp gen	<pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ compgen haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ compgen -c alert egrep fgrep grep l la ll ls if then else elif fi case esac for select while until do done in function time { } ! [[]] coproc __expand_tilde_by_ref __git_eread __git_psi __git_psi_colorize_gitstring </pre>
To know the usage of the command	help or cheat	command -- help	ls -- help	<pre> haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2\$ ls --help List directory contents. Ignore files and directories starting with a '.' by default Usage: ls [OPTION]... [FILE]... Arguments: [paths]... Options: --help Print help information. --format=<format> Set the display format. -C Display the files in columns. -l, --long Display detailed information. -x List entries in rows instead of in columns. -T, --tabsize <COLS> Assume tab stops at each COLS instead of 8 [env: TABSIZE=] -m List entries separated by commas. --zero List entries separated by ASCII NUL characters. -D, --dired generate output designed for Emacs' dired (Directory Edit) --hyperlink[=<WHEN>] hyperlink file names WHEN [default: never] [possible values: never, always, auto] -1 List one file per line. -o Long format without group information. Identical to --format=long with --no-group. -g Long format without owner information. -n, --numeric-uid-gid -l with numeric UIDs and GIDs. --quoting-style <quoting-style> Set quoting style. [possible values: literal, locale, shell, escape] -N, --literal Use literal quoting style. Equivalent to '--quoting-style=literal'. -b, --escape Use escape quoting style. Equivalent to '--quoting-style=shell'. -Q, --quote-name Use C quoting style. Equivalent to '--quoting-style=c'. -q, --hide-control-chars Replace control characters with '?' if they are not escaped. --show-control-chars Show control characters 'as is' if they are not escaped. --time=<field> Show time in '<field>': access time (-u): atime, access, use; change time (-t): ctime, status; modification time: mtime, modification; birth time: birth, creation; -c If the long listing format (e.g., -l, -o) is being used, status change time (the 'ctime' in the inode) instead of time. When explicitly sorting by time (--sort=time or -t) using a long listing format, sort according to the status. -u If the long listing format (e.g., -l, -o) is being used, status access time instead of the modification time. When sorting by time (--sort=time or -t) or when not using a long listing format, sort according to the access time. --hide <PATTERN> do not list implied entries matching shell PATTERN (overridden by -a or -A) </pre>

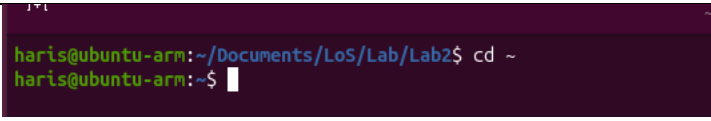
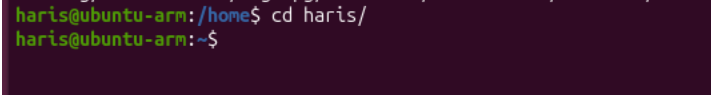
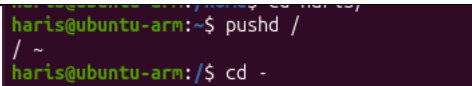
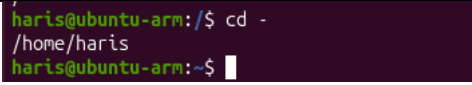
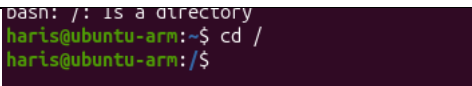
2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date [OPTIONS] [+FORMAT]	date +%A	 <pre> haris@ubuntu-arm:~\$ date +%A Tuesday haris@ubuntu-arm:~\$ date Tue Aug 25 10:25:30 PM IST 2026 haris@ubuntu-arm:~\$ cal 122026 </pre>

To print calendar	cal	cal [OPTIONS] [MONTH] [YEAR]	cal 12 2026	<pre> haris@ubuntu-arm:~\$ cal 12 2026 December 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 </pre>
To print kernel version	uname	uname [OPTIONS]	uname -r	<pre> haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2/Lab2_log_7.0.0-30-generic haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2/Lab2_log_ </pre>
To print default shell	echo \$SHELL	echo \$SHELL	echo \$SHELL	<pre> haris@ubuntu-arm:~\$ echo \$SHELL /bin/bash haris@ubuntu-arm:~\$ </pre>
To print currently logged in user	whoami	whoami	whoami	<pre> haris@ubuntu-arm:~\$ whoami haris haris@ubuntu-arm:~\$ </pre>
To create shortcut for command	alias	alias shortcut_name='command'	alias c='clear'	<pre> haris haris@ubuntu-arm:~\$ alias c='clear' haris@ubuntu-arm:~\$ </pre>
To delete shortcut	unalias	unalias shortcut_name	unalias clear	<pre> haris@ubuntu-arm:~\$ unalias clear bash: unalias: clear: not found haris@ubuntu-arm:~\$ unalias -a haris@ubuntu-arm:~\$ </pre>
To change the timestamp of the file	touch	touch -d "Time/Date" myfile.txt	touch -t 20260820 1530.00 myfile.txt	<pre> haris@ubuntu-arm:~\$ unalias -a haris@ubuntu-arm:~\$ touch -t 202608201530.00 myfile.txt haris@ubuntu-arm:~\$ cd haris@ubuntu-arm:~\$ ls Desktop Downloads Pictures Templates myfile.txt Documents Music Public Videos snap haris@ubuntu-arm:~\$ stat myfile.txt File: myfile.txt size: 0 Blocks: 0 IO Block: Device: 259,2 Inode: 568100 Links: 1 Access: (0664/-rw-rw-r--) Uid: (1000/ haris) Access: 2026-08-20 15:30:00.000000000 +0530 Modify: 2026-08-20 15:30:00.000000000 +0530 Change: 2026-08-25 22:31:22.588027792 +0530 Birth: 2026-08-25 22:31:22.585028161 +0530 haris@ubuntu-arm:~\$ </pre>
To clear the screen	clear	clear	clear	
To create	touch	touch [filename]	touch file1.txt	

empty files				
To know disk usage	du	du [OPTIONS] [directory]	du -h	<pre> haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ du -h 4.0K . haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ </pre>
To know free space in the system	df	df [OPTIONS]	df -h	<pre> haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ df -h Filesystem Size Used Avail Use% Mounted on tmpfs 676M 2.8M 673M 1% /run /dev/nvme0n1p2 29G 11G 17G 41% / tmpfs 1.7G 0 1.7G 0% /dev/shm efivarfs 256K 34K 223K 14% /sys/firmware none 1.0M 0 1.0M 0% /run/credentials none 1.0M 0 1.0M 0% /run/credentials /dev/nvme0n1p1 1.1G 6.6M 1.1G 1% /boot/efi tmpfs 1.7G 8.0K 1.7G 1% /tmp tmpfs 338M 96K 338M 1% /run/user/1000 vmhgfs-fuse 229G 69G 160G 31% /mnt/hgfs haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ </pre>
To know about the Linux release	lsb_release	lsb_release [OPTIONS]	lsb_release -a	<pre> haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ lsb_release -a No LSB modules are available. Distributor ID: Ubuntu Description: Ubuntu 26.04 LTS Release: 26.04 Codename: resolute haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ </pre>
To show login history	last	last [options] [username] [tty]	last -n 10	<pre> haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ last -n 10 haris tty2 local Tue Aug 25 20:25 haris tty2 local Tue Aug 25 20:25 haris tty2 local Tue Aug 25 20:25 wtmpdb begins Tue Aug 25 09:34:48 2026 haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ </pre>
To check the current user privileges	id	id [options] [username]	id	<pre> haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ id uid=1000(haris) gid=1000(haris) groups=1000(haris), dip),46(plugdev),100(users),111(lpadmin),114(lxd) haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ </pre>
To check system uptime	uptime	uptime [options]	uptime -p	<pre> haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ uptime -p up 37 minutes haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2\$ </pre>
To shutdown	shutdown	[sudo] shutdown [options] [time] [message]	sudo shutdown now	
To restart	reboot	[sudo] reboot [options]	sudo reboot	

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd ~	cd ~	
To navigate to the parent directory	cd ..	cd ..	
To navigate to the child directory	cd dir	cd directory_name	
Alternative command to cd	pushd and popd	pushd directory_path	
To go back to the previous directory	cd -	cd -	
To go to the root directory	cd /	cd /	

4. File System

Task	Syntax	Command
How to identify the file system	lsblk -f	lsblk

```

harris@ubuntu-arm:~/Documents/LoS/Lab/Lab2$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                               FSAVAIL FSUSE% MOUNTPOINTS
loop0       squashfs 4.0              0      100% /snap/core24/1588
loop1       squashfs 4.0              0      100% /snap/bare/5
loop2       squashfs 4.0              0      100% /snap/core24/1644
loop3       squashfs 4.0              0      100% /snap/desktop-security-center/149
loop4       squashfs 4.0              0      100% /snap/desktop-security-center/152
loop5       squashfs 4.0              0      100% /snap/firefox/8105
loop6       squashfs 4.0              0      100% /snap/gnome-46-2404/154
loop7       squashfs 4.0              0      100% /snap/mesa-2404/1166
loop8       squashfs 4.0              0      100% /snap/prompting-client/203
loop9       squashfs 4.0              0      100% /snap/prompting-client/221
loop10      squashfs 4.0              0      100% /snap/gtk-common-themes/1535
loop11      squashfs 4.0              0      100% /snap/snap-store/1368
loop12      squashfs 4.0              0      100% /snap/snap-store/1391
loop13      squashfs 4.0              0      100% /snap/snapd/26869
loop14      squashfs 4.0              0      100% /snap/snapd-desktop-integration/363
nvme0n1
├─nvme0n1p1 vfat      FAT32      0020-912D              1G      1% /boot/efi
└─nvme0n1p2 ext4       1.0       3f976e8c-039d-439e-af77-9a6ff16d55db 16G     38% /
harris@ubuntu-arm:~/Documents/LoS/Lab/Lab2$

```

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use **globbing patterns wherever possible**, rather than specifying filenames individually.

- a. List all files ending in .html.

```
ls *.html
```

- b. List all files **other than .html files** using extended globbing.

```
ls !(*.html)
```

- c. List filenames containing at least one numeric digit.

```
ls *[0-9]*
```

- d. List files whose names begin with user followed by a single digit.

```
ls user?.*
```

- e. List files ending in either .txt or .log using a single **extended glob pattern**.

```
ls *.@(txt|log)
```

- f. List files that do **not** end in .txt or .log.

```
ls !(*.txt|*.log)
```

- g. List files beginning with either access or error using extended globbing.

```
ls @(access|error).log
```

- h. List .html files whose filenames contain a numeric digit.

```
ls *[0-9]*.html
```

6. Extended Globbing

- a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

```
ls @(access|error|auth).log
```


Select all files except notes.txt.

```
ls !(notes.txt)
```

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

```
ls auth.log.@(1|2)
```

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log
construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
ls malware*[0-9]*.log
```

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log
use extended globbing to select .log files containing either failed_login or successful_login.

```
ls @(failed_login|successful_login).log
```

Construct a pattern that lists all .log files **except** firewall.log.

```
ls !("firewall.log")
```

7. Filters

a. Display the first **10 entries** of login.log.

```
head -10 login.log
```

b. Display the first **5 entries** of access.log.

```
head -5 access.log
```

c. Display the first **15 security events** from security.log.

```
head -15 security.log
```

d. Display the first **20 events** recorded in incident.log.

```
head -20 incident.log
```

e. Display only the **first entry** in login.log.

```
head -1 login.log
```

f. Display the first **25 web requests** recorded in access.log.

```
head -25 access.log
```

g. Display the last **10 entries** of login.log.

```
tail -10 login.log
```

h. Display the last **5 entries** of access.log.

```
tail -5 access.log
```

i. Display the **20 most recent security events** from security.log.

```
tail -20 security.log
```

j. Display the **25 most recent events** from incident.log.

```
tail -25 incident.log
```

k. Display only the **last entry** in security.log.

```
tail -1 security.log
```

- l. Display the last **50 events** recorded in incident.log.
tail -50 incident.log
- m. Display the contents of login.log starting from **line 41 until the end of the file**.
tail -n +41 login.log
- n. Display the contents of security.log starting from **line 101 until the end of the file**.
tail -n +101 security.log
- o. Find the total number of lines in login.log.
wc -l login.log
- p. Find the total number of lines in access.log.
wc -l access.log
- q. Find the total number of security events in security.log.
wc -l security.log
- r. Determine the total number of events recorded in incident.log.
wc -l incident.log
- s. Count the number of words in login.log.
wc -w login.log
- t. Count the number of bytes in security.log.
wc -c security.log
- u. Display the **line, word, and byte counts** of access.log.
wc access.log
- v. Find the line counts of all four .log files using a single wc command.
wc -l login.log access.log security.log incident.log
- w. Display the first **20 entries** of login.log and, from those entries, display only the **last 5**.
head -20 login.log | tail -5
- x. Display the last **20 entries** of login.log and, from those entries, display only the **first 5**.
tail -20 login.log | head -5
- y. Display the first **30 entries** of access.log and then display only the **last 10 entries** from that output.
head -30 access.log | tail -10
- z. Display the last **30 entries** of access.log and then display only the **first 10 entries**.
- aa. Display the first **50 events** from security.log and then display the **last 10 events** from those 50.
- bb. Display the last **50 events** from security.log and then display the **first 10 events** from those 50.
- cc. Display entries from **line 11 to line 20** of login.log.

- dd. Display entries from **line 21 to line 30** of login.log.
- ee. Display entries from **line 11 to line 20** of access.log.
- ff. Display entries from **line 31 to line 40** of access.log.
- gg. Display security events from **line 21 to line 30** of security.log.
- hh. Display security events from **line 51 to line 60** of security.log.
- ii. Display incident events from **line 101 to line 110** of incident.log.
sed -n '101,110p' incident.log
- jj. Display incident events from **line 201 to line 220** of incident.log.
- kk. Display incident events from **line 491 to line 500**.
- ll. Take the first **20 entries** of login.log and count how many lines are displayed.
head -20 login.log | wc -l
- mm. Take the last **15 entries** of access.log and count how many lines are displayed.
tail -16 access.log | wc -l
- nn. Take the first **50 entries** of security.log and count the number of words in those entries.
- oo. Take the last **25 entries** of security.log and count the number of lines.
- pp. Extract lines **21–30** from login.log and count the number of extracted lines.
sed -e '21,30p' login.log | wc -l
- qq. Extract lines **51–60** from access.log and count the number of extracted lines.
sed -n '51,60p' login.log | wc -l
- rr. Extract events **101–150** from incident.log and count how many events are displayed.
- ss. Display the first **100 events** from incident.log, take the last **20 events** from that output, and count the number of lines.
head -100 incident.log | tail -20 | wc -l